A Conformable Moments-Based Deep Learning System for Forged Handwriting Detection

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***Abstract*— Detecting forged handwriting is important in a wide variety of machine learning applications, and it is challenging when the input images are degraded with noise and blur. This article presents a new model based on conformable moments (CMs) and deep ensemble neural networks (DENNs) for forged handwriting detection in noisy and blurry environments. Since CMs involve fractional calculus with the ability to model nonlinearities and geometrical moments as well as preserving spatial relationships between pixels, fine details in images are preserved. This motivates us to introduce a DENN classifier, which integrates stenographic kernels and spatial features to classify input images as normal (original, clean images), altered (handwriting changed through copy-paste and insertion operations), noisy (added noise to original image), blurred (added blur to original image), altered-noise (noise is added to the altered image), and altered-blurred (blur is added to the altered image). To evaluate our model, we use a newly introduced dataset, which comprises handwritten words altered at the character level, as well as several standard datasets, namely ACPR 2019, ICPR 2018-FDC, and the IMEI dataset. The first two of these datasets include handwriting samples that are altered at the character and word levels, and the third dataset comprises forged International Mobile Equipment Identity (IMEI) numbers. Experimental results demonstrate that the proposed method outperforms the existing methods in terms of classification rate.**

***Index Terms*— Conformable moments (CMs), convolutional neural networks (CNNs), deep learning, forged handwriting, fractional calculus, handwritten documents.**

I. Introduction

IT IS noted from previous work [1] that developing systems for detecting suspicious activities is gaining popularity in the field of machine learning and forensic security applications [2], [3]. This indicates that the alteration or tampering of content in images is a serious threat for forensic applications, such as forging property, insurance, certificates, banking documents, and other sensitive documents such as creating fake suicide notes and fake answer scripts. In addition, there are several occasions, such as gender, writer, nationality identification, age estimation, behavior identification using handwriting analysis, which can assist forensic investigation teams to identify criminals. In these situations, altering, tampering and forging of content can have tragic consequences, and it may even cause death of a person. Therefore, robust forged handwriting detection, which is invariant to noise and blur, is essential.

There are methods available for detecting forgeries in document images, video, and general images [4], [5], [6], [7], [8], [9], [10], [11], [12]. Most of the methods rely on image features such as texture, color, and distortion at the edges for detecting forgeries in the images. However, these features may not be adequate for detecting forgeries in handwritten document images due to large variations in the writing [13], [14]. To overcome this limitation, a few methods have been developed that work based on inconsistency and irregular patterns of forged texts [15], [16]. But these techniques are not robust to handwritten documents affected by noise and blur because of inconsistencies and irregular patterns, which are used as a cue for forgery detection.

Moreover, it can be observed that handwritten documents are comprised of words affected by noise, blur, text with poor quality, and degradations due to errors in devices, flaws in the process of image acquisition, paper quality, and camera movements while capturing images. The factors make the problem of forged handwritten text detection in noisy and blurred environments complex and challenging. In addition, the forgery at a character level adds more complexity to the problem. This is because character-level forgeries do not provide noticeable differences compared to word and text lines forgery for detection. Thus, there is a need for addressing the

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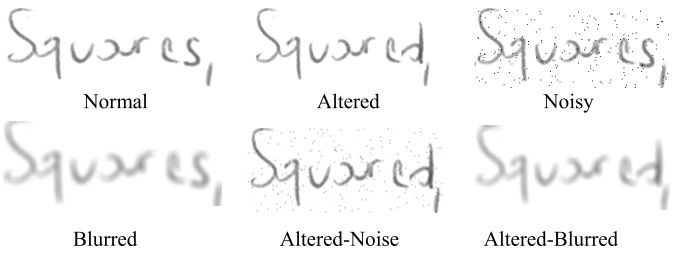


Fig. 1. Examples of sample images of the six classes.

challenges of forged handwritten text detection in noisy and blurry environment.

Therefore, we define six classes, namely: 1) Normal words that are not affected by noise or alterations; 2) Altered words created through standard copy-paste and insertion operations; 3) Noisy words where different noise densities are added to the original image; 4) Blurred words where the original image is blurred to various levels; 5) Altered-Noisy words created by adding noise to previously altered images of handwriting; and 6) Altered-Blurred words created by adding blur effects to handwriting that has been altered. For example, the images for the above six classes can be seen in Fig. 1, where the naked eye can easily detect blurred and noisy words but has more difficulty in identifying words that have been altered. Overall, to the best of our knowledge, this is the first work focusing on forged handwriting detection in noisy and blurred environments.

The key contributions of the proposed work are as follows: 1) conformable moments (CMs) are employed here for enhancing the finer details in images irrespective of degradations introduced by forgeries, noise, and blur; 2) a new deep learning model is developed by exploring the combination of stenographic kernels and spatial filters; 3) to the best of our knowledge, the existing forgery detection methods have not explicitly considered noisy and blurred input images. Thus, this work is new in this respect; and 4) since it is the first of its kind, the creation of a dataset with its ground truth for forgery detection is a cumbersome task. We have done this, and we shall release the code and dataset publicly to support research reproducibility.

The remainder of this work is structured as follows. A review of the related methods of fraudulent document identification, source printer identification, and forged text detection is presented in Section II. Section III introduces a new model of CMs for enhancing the finer details in the input images and deep ensemble neural networks (DENNs) for detection of forged handwritten text. To validate the proposed method, experimental analysis on different benchmark datasets is discussed in Section IV. Lastly, Section V summarizes the proposed work.

II. Related Work

In the past decade, we can see extensive research being undertaken for fraudulent document identification, source printer identification, and forged text detection in document images. The methods can be classified into two broad categories, namely genuine document identification and printer identification.

Loc *et al.* [17] proposed an approach for securing genuine documents by exploring adversarial networks. It embeds watermarking to hide secret information in the document. However, the method is sensitive to distorted documents. Berenguel *et al.* [18] proposed a method for detecting counterfeit documents based on a deep learning model. Due to the use of high-quality printers, fraudulent documents were expected that did not leave any noticeable mistakes. However, the work assumed that when a counterfeit document is created, some abrupt changes in the background texture of the document would be evident. Raghunandan *et al.* [4] aim to identify fraudulent handwritten documents by considering them as a two-class problem. The approach uses Fourier coefficients for studying the quality of handwritten documents. Quality-based features may not be robust for documents affected by adverse factors, such as distortions, noise, blur, and forgery operations.

The above review on fraudulent document identification methods show that most approaches use documents that do not suffer from degradations, poor quality, and aging for forgery detection. This is because the methods work on the assumption that the forgery operations introduce distortions. This assumption may not be true for documents with noise, blur, and degradations.

In a similar way, there is another set of methods that identifies the source printer for detecting forged documents. Wang *et al.* [12] propose a method for printer identification based on different texture features and passing extracted features to an SVM for the classification of documents printed by different printers. Bibi *et al.* [19] focus on developing a method that identifies the source printer, which is text independent. To achieve this, the approach uses deep convolutional neural networks (CNNs) in contrast to hand-crafted features. Joshi and Khanna [20] propose a method for source printer classification by defining local texture descriptors. However, the extracted features are sensitive to document aging and degradation. One can note from the review on the methods of source printer identification that none of the methods uses handwritten documents for altered text detection. This is because the methods expect unique noise or distortions created by different printers.

Recently, Shivakumara *et al.* [21] targeted forged International Mobile Equipment Identity (IMEI) number detection by exploring color features and fusion concepts. Similar to [21], Nandanwar *et al.* [16] propose an altered text detection method in document images. It considers PDF documents for altered text detection. The method uses the input image and the image reconstructed by DCT for finding differences between input and reconstructed images. To address the challenges of forged handwritten documents in noisy and blurry environments, Kundu *et al.* [15] proposed a method based on pixel distributions corresponding to the Fourier spectrum and its shape. Although the method is robust to blur and noisy words, it does not perform well for character-level forgeries and altered-noise, blurry images. Megahed *et al.* [22] proposed a method for forgery detection in handwritten documents based on ink color features. These features are extracted

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from the R, G, and B color spaces for differentiating text written by different writers. Nandanwar *et al.* [23] proposed a method for forged international mobile equipment number and altered air ticket detection employing DCT phase-based features. However, these methods were developed assuming clean input images and may not work well for images affected by noise and blur.

Considering the above review, most existing methods do not perform well for documents that suffer from poor quality, degradation, and distortions because distortions created by forgery operations create confusion with the actual distortion of documents. Furthermore, variations in handwriting make the problem more challenging, and to the best of our knowledge, these are still considered as open issues. Hence, inspired by the special properties of CMs, which can preserve salient information, such as fine edges irrespective of distortion, noise, and blur [24], [25], we explore CMs for preserving structures of characters in noisy and blurry environments. This is promising because CMs involve fractional calculus which can solve the nonlinear complexity problem [26], [27], and geometric moments which help to define a spatial relationship between pixels. Since CMs assist in finding the finer details in noisy and blurry environments, it results in unique edge patterns for images of different classes. This motivated us to propose a new architecture based on stenographical kernels and spatial filters [28] for extracting such unique patterns.

III. Proposed Method

Since there are several powerful word segmentation approaches that work well irrespective of script, orientation, degradation, and distortion [29], [30], [31], we consider words as the inputs for the purpose of altered text detection. The proposed work considers six classes as discussed in the introduction section namely: 1) Normal words; 2) Altered words; 3) Noisy words; 4) Blurred words; 5) Altered-Noisy words; and 6) Altered-Blurred words.

Inspired by the fact that CMs can find solutions to complex problems such as nonlinear complexities [24], [25], we explore CMs for altered text detection. It is expected that CMs provide enhanced images for the six classes by preserving the finer details [26], [27] of characters. This results in the unique edge patterns for all the six classes, which can be used for the classification of altered words from normal, noisy, and blurred words. Motivated by stenographical kernels and spatial kernels, which help us to extract contextual information with the help of local spatial information, we propose to design a DENN. This approach integrates stenographic residual features, and features extracted by the CNN from photograph montages for detecting altered images from other images.

The proposed method extracts statistical features using Stenographic Residuals with High Pass Filters for both input and enhanced images given by CMs. At the same time, to increase the strength of stenographic residual features, we generate a photograph montage using inputs and CM images from which the proposed method extracts Spatial CNN features using the EfficientNet [32] backbone trained on the ImageNet dataset as a feature extractor. The motivation

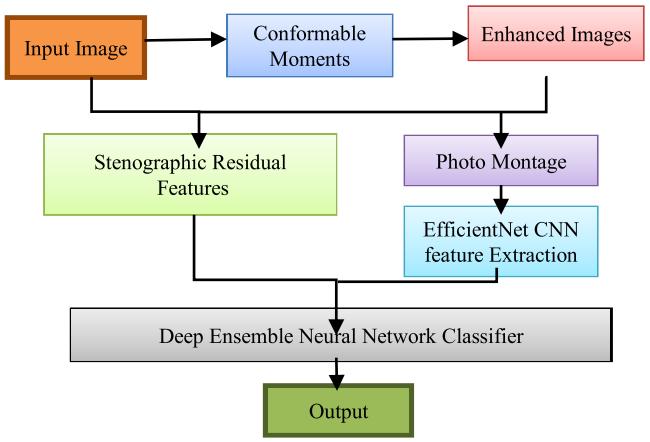


Fig. 2. Framework of the proposed method.

to propose EfficientNet over Densenet, Resnet, etc., is that EfficientNet is designed through a neural architecture search for getting the best combination of blocks and layers. It is based on a model scaling method using effective compound coefficients, which means we can increase the model size as per the requirements and needs. It has a low number of parameters compared to other SOTA methods and has high accuracy, which implies less training time and an improvement in real-time performance. The features are then fed to a DENN classifier, which consists of two sets of classifiers for Stenographic features and CNN features, along with the fusion layer for classification. The framework of the proposed method can be seen in Fig. 2.

A. Theory of CMs

As we have discussed earlier, when we perform forgery operations on an input image in noisy and blurry environments, the quality of the images degrades. When the quality degrades, a variance increases between the distortion created by the forgery operation, and the noisy, blurry handwriting variations. Therefore, to cope with this challenge, we introduce CMs to improve the quality of the image, spatially. In the case of CMs, the basis function with an alpha power is important to enhance the edge pixel values. It is stated due to Anderson and Ulness [25] that the CMs are derived from Conformable Calculus (CC) based on control theory to estimate the performance of the proportional-differentiation controllers in terms of a well-known error function. We explore this idea to formulate a new CM and derive interesting results by establishing a relationship between the class of CC and control theory and use it for altered text detection. We briefly present a summary of CMs here to convey the idea of the operation using CC. Details about CC can be found in [25].

Overall, fractional calculus reduces the influence of backgrounds and distortion, while moments preserve spatial information of the finer details (edges) in the images such that one can expect unique patterns for each of the classes.

As per [25], considering ψ_1, ψ_0 as the fractional functions of fine-tuning quantities and I as a differentiable function and I' as the first derivative of I, conformable derivative can be

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formulated in the subsequent form

$$\Delta^{\alpha} I(x) = \psi_1(\alpha, x) I(x) + \psi_0(\alpha, x) I'(x) \tag{1}$$

for the following limits of the functions $\psi_1 : [0, 1]XR \rightarrow [0, \infty)$ and $\psi_0 : [0, 1]XR \rightarrow [0, \infty)$

$$\lim_{\alpha \to 0} \psi_0(\alpha, x) = 0, \quad \lim_{\alpha \to 1} \psi_0(\alpha, x) = 1$$
$$\lim_{\alpha \to 0} \psi_1(\alpha, x) = 1, \quad \lim_{\alpha \to 1} \psi_1(\alpha, x) = 0.$$

To reach the above-mentioned definitions, we will introduce the following cases in view of our approach:

$$\psi_0(\alpha, x) = \alpha x^{1-\alpha}, \quad \psi_1(\alpha, x) = (1 - \alpha) x^{\alpha}. \tag{2}$$

The CM-generating function $\widehat{M_X}(t), t \in \mathrm{R}$ of any random variable X is defined by

$$\widehat{M_X}(t) = \Xi[e_{\lambda}(Xt, 0; \alpha)] := \Xi[e_{\lambda}(Xt; \alpha)]$$

where

$$e_{\lambda}(Xt; \alpha) = \exp\left(\int_0^{Xt} \frac{\lambda - \psi_1(\alpha, x)}{\psi_0(\alpha, x)} dx \right). \tag{3}$$

Substituting (2) in (3), we have

$$e_{\lambda}(Xt; \alpha) = \exp\left(\frac{\lambda}{\alpha^2} (Xt)^{\alpha} - \frac{(1 - \alpha)}{2\alpha^2} (Xt)^{2\alpha} \right). \tag{4}$$

The discrete behavior of the (x, y) can be studied by dividing it into pixels having a discrete intensity level. Two-dimensional moments of a digitally sampled $M \times N$ image that has gray function $f(x, y)$, $(x, y = 0, \ldots, M - 1)$, is given as

$$\widehat{\mathfrak{M}_{pq}} = \sum_{x=1}^{M} \sum_{y=1}^{N} \Lambda_{\lambda, pq}(x, y; \alpha) \cdot I(x, y) \tag{5}$$

where $M \times N$ is the size of the image, $\Lambda_{\lambda, pq}(x, y; \alpha)$ is the basis function of the discrete formula, and $I(x, y)$ is the intensity of a pixel in the digital feature at the xth row and yth column.

To be consistent with the application of this study, the power term (2α) of the exponential function in (4) has been approximated into (α) by using the local linear approximation [33].

Hence, we approximate the basis function as follows:

$$\begin{aligned} \Lambda_{\lambda, pq}(x, y; \alpha) &= \left(\frac{\lambda}{\alpha^2} (x^{\alpha})^p (y^{\alpha})^q - \frac{(1 - \alpha)}{2\alpha^2} (x^{\alpha})^p (y^{\alpha})^q \right) \\ &= \left(\frac{2\lambda - (1 - \alpha)}{2\alpha^2} \right) (x^{\alpha})^p (y^{\alpha})^q \\ &= \left(\frac{\lambda - 0.5(1 - \alpha)}{\alpha^2} \right) (x^{\alpha})^p (y^{\alpha})^q, \ \alpha \neq 0. \end{aligned} \tag{6}$$

Consequently, using (6), (5) can be written as follows:

$$\widehat{\mathfrak{M}_{pq}} = \left(\frac{\lambda - 0.5(1 - \alpha)}{\alpha^2} \right) \sum_{x} \sum_{y} (x^{\alpha})^p (y^{\alpha})^q \cdot I(x, y) \tag{7}$$

where $(x^{\alpha})^p (y^{\alpha})^q$ is the basis function, and $\lambda > 1/2(1 - \alpha)$.

Equation (7) defines CMs, which are used to enhance the input image. The main advantage of CMs lies in their ability to accurately describe the information contained in the image features, which makes them an efficient image-enhancing algorithm in the case of degraded images. In the proposed CM image enhancement model, the key parameter is α, where the performance of the CM basis function of the α power is utilized to enhance the illumination value of the pixels of the image, which might influence the accuracy of the detection process.

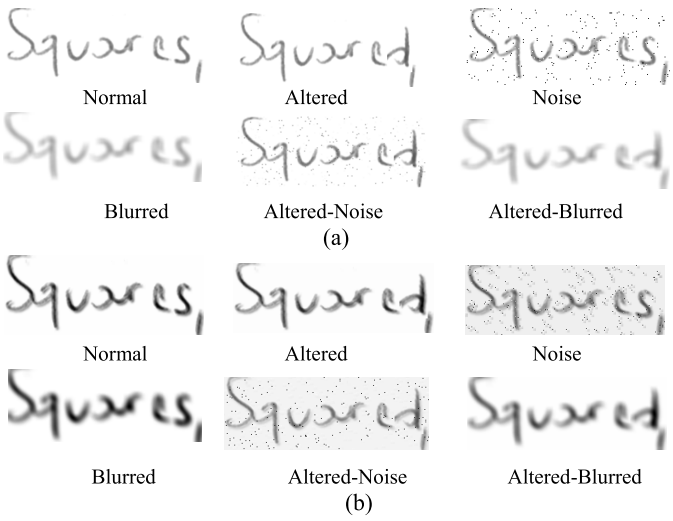


Fig. 3. Examples of the enhanced images by the proposed CMs for the input images of six classes. (a) Input images of six classes and (b) result of CMs.

TABLE I

QUALITY MEASURES FOR INPUT AND ENHANCED IMAGES OF ALL THE SIX CLASSES (N: NIQUE, B: BRISQUE, P: PSNR, AND S: SSIM)

Input Images											
Normal		Altered		Noise		Blurred		Altered-Noisy		Altered-Blurred	
N	B	N	B	N	B	N	B	N	B	N	B
15.1	46.5	13.9	47.1	26.5	49.3	15.2	47.7	21.0	48.5	13.8	48.2
Enhanced images											
13.2	47.9	12.8	48.3	23.5	46.8	14.0	45.9	19.9	45.1	12.7	46.8
PSNR and SSIM using input image normal class and enhanced images											
P	S	P	S	P	S	P	S	P	S	P	S
23.9	0.85	23.1	0.87	22.5	0.80	21.9	0.81	21.9	0.82	21.3	0.80

The effect of the proposed CMs for sample images of the six classes, shown in Fig. 3(a), can be seen in Fig. 3(b). It is noted from Fig. 3(a) that one does not notice abrupt changes or inconsistency patterns caused by forgery operations in altered images compared to the normal ones. However, from the results of CMs shown in Fig. 3(b), a greater level of brightness can be observed for the normal words, while there is a fainter level for the altered words. At the same time, when we observe the input and enhanced images affected by noise and blur in Fig. 3(a) and (b), it can be noted that the brightness is increased for enhanced images compared to the input images. Due to the brightness in CM images, the structures of characters can be preserved and can be extracted, despite the fact that the images are affected by noise and blur.

To demonstrate how CMs help in preserving structural details and spatial information for the altered images through statistical analysis, we estimate different measures namely, NIQUE, Blind Reference less Image Spatial Quality Evaluator (BRISQUE), PSNR, and SSIM for the images shown in Fig. 3. The results are reported in Table I. It is evident from

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Algorithm 1 CMs

1: Initialization: I=Input image, set the fractional power $\alpha = 0.6$
2: **For** each Input **do**
3: $\{B_1, B_2, \ldots, B_n\} \leftarrow$split I into non-overlapped blocks size of 4×4 pixels
4: **For** $i = 1$ to n **do**
5: $M_{pq} \leftarrow I$ // Conformable Moments is calculated for each block of pixels as defined in Equation (7), where i denotes the ith block of 4×4 dimensions.
6: **EndFor**
7: Calculate the sum of the M_{pq} for all image blocks
8: **EndFor**
9: Display enhanced image

Table I that the NIQUE and BRISQUE are reporting low values for the enhanced images of all the classes compared to the input images, except for the altered class for which only BRISQUE does not score low values. The low values of NIQUE and BRISQUE indicate better quality. To verify the same observation, we also estimate PSNR and SSIM for the images shown in Fig. 3. The reason to consider PSNR and SSIM is that these two measures are defined exclusively for measuring pixel signal-to-noise ratios and the structural similarity index between input and enhanced images. For this experiment, we considered input images of the normal class as a ground truth for calculating measures; and the results are reported in Table I. The PSNR and SSIM are calculated by comparing enhanced images of all the classes with the input image of the original class. In contrast to NIQUE and BRISQUE, high values of PSNR and SSIM indicate better quality. Therefore, Table I illustrates that the PSNR and SSIM score almost similar values for all the classes, compared to the normal class. With the results of NIQUE, BRISQUE, PSNR, and SSIM, one can infer that the proposed CMs preserve the finer details and spatial information for the altered images, which facilitates successful classification of altered images.

The method divides the whole image into a number of nonoverlapping blocks with size of 4×4, and then it extracts CMs for each block as defined in (7). The steps for obtaining enhanced images using the proposed CMs are presented in the Algorithm below. The block size of 4×4 is determined empirically.

B. DENN for Classification of Altered Handwriting

In the case of handwriting, pixel values and their neighboring values play a key role in defining a spatial relationship between pixels, which can be used to extract structures of characters. In our work, the altered images affected by forgery operations, noise, and blurred images, are considered for classification, which affect the structures of characters. At the same time, since a normal word is not affected by any forgery operations, noise, and blur, the character structures are not affected, and hence the word is not affected. In this work, it is expected that forgery at the character level, and the alteration caused by forgery operations, may not be noticeable. These are the observations for classifying images. Inspired by the special properties of Stenographic kernels that are popular for extracting the spatial dependency of pixels, which can be named as spatial context features, we explore stenographic kernels through residuals using different orders of high-pass filters [28]. We extract stenographic features by merging several smaller residuals of high-pass filters and then concatenating all the residuals to obtain the feature vector. Therefore, stenographic features are the combination of sub-models known as residuals.

The stenographic residual sub-model $\mathbf{R}$ is computed from noise residuals, as defined as follows:

$$\mathbf{R} = \left(I_{ij}\right) \in \mathbb{R}^{n_1 \times n_2}. \tag{8}$$

Based on high pass filters as described in [25], we extract stenographic features as defined below:

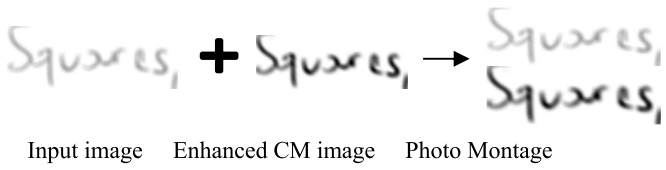
$$I_{ij} = \hat{X}_{ij}\left(\mathcal{N}_{ij}\right) - cX_{ij} \tag{9}$$

where $c \in \mathbb{N}$ is a residual order, $\mathcal{N}_{ij}$ is a local neighborhood of the pixel X_{ij}, $X_{ij} \notin \mathcal{N}_{ij}$ and $\hat{X}_{ij}(.)$ is a predictor of cX_{ij} defined on $\mathcal{N}_{ij}$. The $\{X_{ij} + \mathcal{N}_{ij}\}$ set is called the support of the residual. The advantage of modeling residual instead of pixel values is that image content is largely suppressed in $\mathbf{R}$, which has a much narrower dynamic range, thus allowing a more compact and robust statistical description. All residuals are built as locally supported linear filters, whose outputs are possibly combined using minimum and maximum operators to increase their diversity. The residuals are divided into six classes, depending on the central pixel predictor. The classes are labeled as 1st, 2nd, 3rd, SQUARE, EDGE3 $\times$ 3, and EDGE5 $\times$ 5. The description of all the 39 residuals can be obtained in [28]. The proposed method calculates all the 39 residuals to extract stenographic features of different dimensions for both the input and enhanced images given by CMs. This results in a feature vector of length 12 753, and then the feature vectors are concatenated into a single dimension vector of size $n = 25\,506$.

It is true that each pixel value contributes to defining character structures in handwriting images. For example, intensity values are almost the same for pixels of each character in word images. Therefore, to strengthen the stenographic features, motivated by CNNs, which demonstrate promising performance for image classification [32], we explore CNNs for extracting features by feeding the images as inputs. The proposed method uses the EfficientNet neural network architecture, trained on a large ImageNet dataset without the top classifier, which acts as a backbone feature extractor in this work. The extracted features are then passed through the GlobalMP layers to obtain the feature vector of a single dimension size, f. In this work, the proposed method passes a montage of input images and CM images as shown in Fig. 4 to the EfficientNet CNN backbone to extract convolutional features.

To analyze the contribution of stenographic features, and the features of photograph montage images, we calculate the standard deviations for feature vectors given by the stenographic residuals of different kernels. The feature matrix contains 39 feature vectors extracted from input images, and 39 more feature vectors extracted from CM images, which results in

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Input image Enhanced CM image Photo Montage

Fig. 4. Photograph montage creation using input and enhanced CM images.

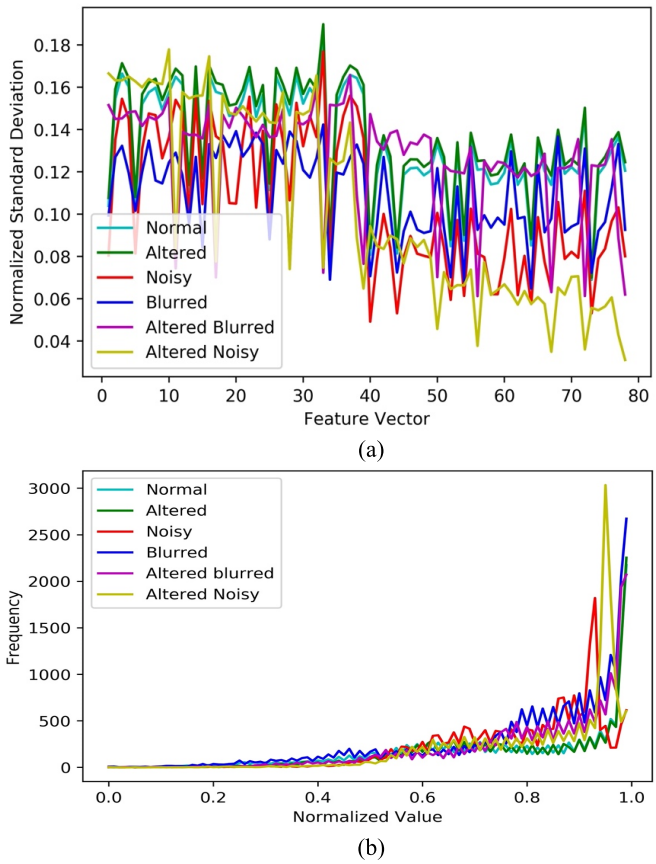


Fig. 5. Effects of stenographic features and the histogram of photograph montage images for classification. (a) Line graph for the standard deviation of stenographic features for the respective six classes. (b) Histogram for Photograph Montage images of six classes.

78 feature vectors. The line graphs for the standard deviations of the respective 78 feature matrices of the six classes are illustrated in Fig. 5(a), where we can see differences in the line graph behavior for all the six images in terms of variations. This indicates that the stenographic features contribute to the image classification of the six classes. Similarly, for the pixel values of the photograph montage images of six classes, the proposed method performs a histogram operation, as shown in Fig. 5(b). It is noted from Fig. 5(b) that the shapes of the histograms are different for all the six images. Therefore, one can assert that the features of the photograph montage images are effective in differentiating the six classes. Overall, we can conclude that the features extracted from stenographic residuals and the features extracted from the photograph montage images have the ability to differentiate the images of six classes, and hence both sets of features contribute equally toward classification.

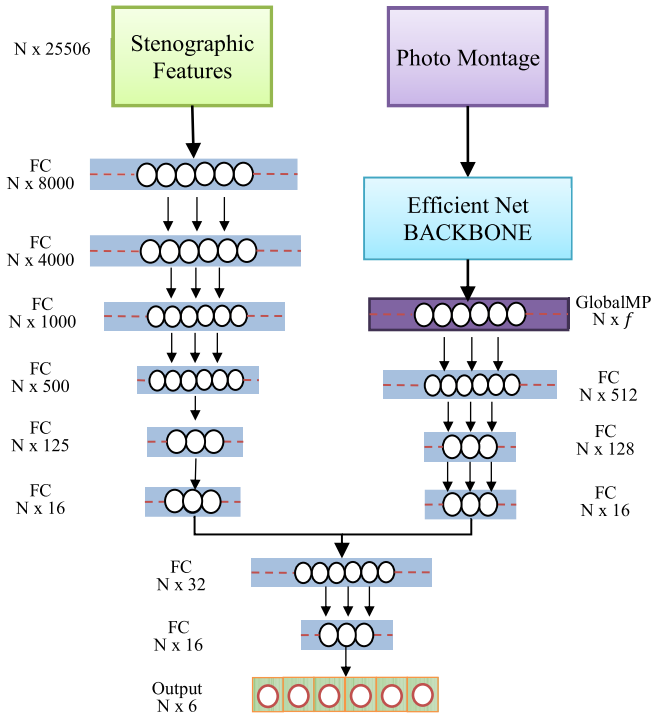


Fig. 6. DENN Architecture.

To integrate the stenographic features and CNN-based features, we propose a DENN Architecture as shown in Fig. 6, where one can see how the proposed method combines both the features. The stenographic features extracted from input and CM images results in a feature vector of size n, which is then passed through the fully connected neural network (F-CNN-1). It consists of six fully connected (FC) layers. The final layer of this FC network contains 16 neurons. Conversely, the features extracted using the efficient net CNN backbone from the montage of inputs and CM images are fed to a fully connected neural network (F-CNN-2) having global max pooling (MP) and three FC layers. The final layer of F-CNN-2 has 16 neurons. The DENN fuses the outputs of F-CNN-1 and -2 and subsequently passes through the FC layer to the final output layer (multilabel classification) containing six neurons, one for each of the six labels (i.e., normal, altered, noisy, blurred, altered noisy, and altered blurred). This architecture uses the "ReLU" activation function for all the layers except for the output layer, where we use the "Softmax" [34] activation function. With "ADAM" [35] as an optimizer and "Categorical Cross Entropy" [36] as the loss function (ℓ), which is defined in (9), where p represents the labels and $\hat{p}$ are the predicted probabilities for the respective C classes used in this work. The dropout rate of 0.2 is added in between the layers to reduce over fitting, which also results in the generalization of our model

$$\ell(p,\hat{p}) = -\sum_{i=1}^{C} p_i \log(\hat{p}_i) \tag{10}$$

where C is the number of classes.

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We use the cyclical learning rate (CLR) [37] strategy to train the proposed architecture. CLR dramatically reduces the number of experiments required for tuning and finds an optimal learning rate. This strategy allows the learning rate to oscillate back and forth between upper and lower bounds when training, slowly increasing and decreasing the learning rate after every batch update. This leads to an optimal way of improving model performance. To find the upper and lower bounds, we have utilized the same approach as defined in CLR, where we train our network for a small number of epochs with a very low learning rate, such as 1e-10, which starts increasing our learning rate exponentially after each batch update. Training will continue until we reach a very high learning rate (1e + 01).

Testing the plot of smoothed loss over time enables us to determine the lower and upper bounds of the learning rate. When the learning rate is large enough for the loss to decrease, it is considered as a lower bound of the learning rate. When the learning rate is too large to the point where the loss starts to increase, it is considered as an upper bound of the learning rate. For this work, the upper bound of the learning rate is determined to be 5e-2 and the lower bound is found to be 1e-5. Then the proposed method employs CLR for 96 epochs for a step size of 8 epochs and a batch size of $N = 16$ images, leading to 6 full CLR cycles to oscillate back and forth between upper and lower bounds. For all the experiments, we use a system with the Nvidia Quadro M5000 GPU for training and testing of the architecture and a python framework along with TensorFlow-v2.1 for this application. The datasets are divided into 80% and 20% for training and testing for all the experiments in this work. The parameters used in this work, namely image size, the number of FC layers, neurons, batch size, learning rate, etc., used in this work are determined experimentally. The system takes approximately 1.5 min for each epoch on the proposed dataset, for a total of 2 h 20 min to complete all the experiments.

To analyze the time complexity of the proposed method, we considered the following key steps: 1) CMs enhanced image; 2) stenographic feature extraction; and 3) deep CNN ensemble for classification. Since the parameters of the CMs depend on the number of overlaps in the blocks of $a \times a$ pixels, the time complexity for an image is $O(n)$ in the worst case, and the same for the best case, which is $\Omega(n)$. It is noted that the time complexity of stenographic feature extraction for an image is $O(n^2)$, where n is the number of pixels in the image. Therefore, the time complexity for extracting Stenographic features from the image is $O(n^2)$ in the best case, and $O(n^3)$ in the worst case. In addition, the time complexity for the classification step using a Deep CNN ensemble is $\Omega(n)$. In summary, the overall time complexity of the proposed method in the best case is $O(n) + O(n^2) + O(n) \Leftrightarrow O(n^2)$, and in the worst case is $O(n) + O(n^3) + O(n) \Leftrightarrow O(n^3)$.

IV. Experimental Results

There is no publicly available standard dataset for detecting altered handwriting at the character level in noisy and blurry environments. In this work, we created a new dataset for evaluating the proposed method. We use the standard forgery operations, namely, copy-paste which copies characters either from the same document or different documents to paste them to target locations in the original image. The insertion operation inserts characters at appropriate locations in the original image using Microsoft Paint. The process we used in creating our dataset can be seen in Fig. 7, where (a) shows how a character "e" is inserted at the end of a word to obtain an altered word. In the same way, the snapshots in (b) and (e) show the steps for performing a copy-paste operation. For an original image shown in Fig. 7(b), the character to be inserted in the original image is chosen from a different image as shown in Fig. 7(c). The chosen character is cropped manually and carefully as shown in Fig. 7(d), and then the cropped character is pasted at the beginning of the original image as shown in Fig. 7(e), resulting in a word altered through a copy-paste operation.

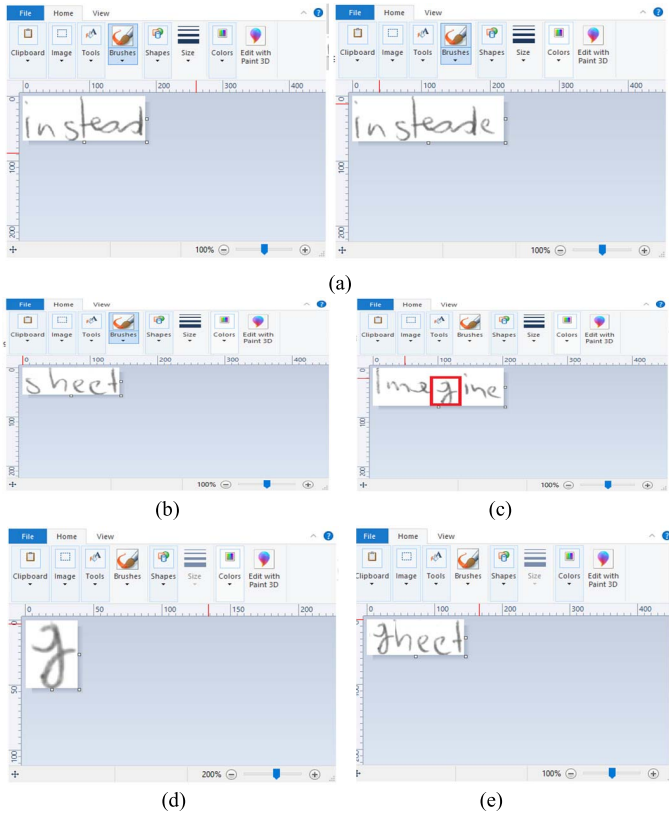


Fig. 7. Procedure and operations involved in creating altered text for our dataset. (For better, use 300%.) (a) For original text, we manually inserted the character "e" using the brush stroke of the same color as that of source image, (b) source image to be altered, (c) character to be inserted, (d) cropping character, and (e) altered text using copy-paste.

This is the approach we used to create our dataset of altered handwritten images. To test our proposed method for detecting alterations in noisy and blurred environments, we used Gaussian functions to add different levels of noise and blur to the original and altered images, yielding four more classes, namely: noisy, blurred, altered-noisy, and altered-blurred. We added noise and blur to the entire image in each case [38], [39], [40] as is standard practice when testing for

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robustness. The original, clean images are designated as the Normal class. Therefore, in total, six classes are employed for our experiments in this work. Each class consists of 500 images, hence there is a total of 3000 images. Since we use datasets originally developed for writer identification and word spotting research [41], one can expect large variations in the handwriting. This makes the data even more challenging for our work. Note that the code and the dataset developed for this work will both be made freely available.

To test the effectiveness of the proposed method, we also considered the benchmark datasets for forged word detection created by performing forgeries at the word level, in this case, the ACPR 2019 dataset [15]. This dataset consists of four classes as in the proposed work, namely: normal, forged, noisy, and blurred images. However, the forged words are created by tampering the whole word in contrast to our dataset, where forgeries are performed at the character level. However, for noisy and blurred images creation, the same operations are used as in the proposed work. As a result, the proposed dataset is much more challenging for altered handwriting detection compared to the ACPR 2019 dataset. The dataset comprises 800 images, which includes 200 images for each class.

There is one further benchmark dataset created for detecting altered prices on receipts. It is published in the ICPR 2018 Fraud Contest Data (FCD) report [42]. Most of the receipts have numerals of two-five digits, date-time, and some character text, which includes forged digits and characters. This dataset is different from ours and the ACPR 2019 datasets because a receipt contains printed or typed digits but not handwriting. In addition, distortions created by forgery operations are noticeable as compared to ours and the ACPR 2019 datasets. However, the challenge of this dataset is that the length of the digits can be shorter than that of handwritten words considered in ours and the ACPR 2019 datasets. The dataset includes 527 originals and 302 forged images, which results in a total of 829 images for experimentation.

The above-mentioned datasets are created for detecting forged text in document images, but not images with a cluttered background. To test the proposed method for forged text detection in cluttered background images, we considered one further standard dataset, namely, the forged IMEI number dataset [21]. The forged IMEI numbers are created by altering 2–3 numerals either by copy-paste or insertion operations. The main challenge of this dataset is that the numerals are embedded in mobile images, and hence it is expected that these are more complex when compared to document images where we can see plain backgrounds. This dataset consists of 500 original images and 500 forged IMEI number images, which gives a total of 1000 images for experimentation.

In summary, we consider four datasets, including our dataset, ACPR-2019, ICPR 2018-FDC, and IMEI number datasets, which give 3000, 800, 829, and 1000 images, respectively, for evaluating the proposed method. In total, 5629 images are considered for experimentation. Since all the four datasets are different in nature and challenges, we believe that the datasets considered are substantial enough to demonstrate that the proposed method is application-independent, and it can detect altered images at the word and the character levels in noisy, and blurred environments.

The proposed work consists of two stages, namely enhancement by CMs, and altered handwriting detection by a DENN. To demonstrate that the proposed enhancement is effective and useful, we implemented three existing enhancement methods, which were developed for improving poor-quality images. These include Raghunandan *et al.*'s [27] method, which proposes fractional calculus for enhancing the poor quality of license plate images, Shamasneh *et al.*'s [26] method, which proposes different fractional calculus for enhancing the poor quality of Kidney images, and Chowdhury *et al.*'s [43] method, which exploits a deep learning model for enhancing degraded license plate images. We chose the above three methods for a comparative study because the primary objective of these existing methods is to improve poor-quality images. The methods [26], [27] use fractional calculus for addressing the issues of poor quality. To show that fractional calculus alone is not sufficient to handle the challenges of altered text detection in handwritten images, we compared the proposed method with the existing methods. Similarly, the method [43] which exploits deep learning models is popular for finding solutions to complex issues and is compared to demonstrate that a deep learning model alone is not adequate to improve the quality of handwritten images in noisy and blurry environments.

For altered handwriting detection, to show that the proposed method is effective and useful, we implemented the following existing methods for a comparative study in this work. Wang *et al.* [12] proposed texture features and a support vector machine classifier for printer source identification. Shivakumara *et al.* [21] used a fusion approach for forged IMEI number detection. Kundu *et al.* [15] proposed Fourier spectral density for forged handwritten word detection in noisy and blurry environments. Megahed *et al.* [22] proposed ink color-based features for handwriting forgery detection. Nandanwar *et al.* [23] explored DCT-phase statistics for altered air ticket detection. The reason to choose these methods for our comparative study is that they share a similar aim: detecting forgeries. For Megahed *et al.* [22], we extracted features as defined in the method, but for classification we used a linear neural network classifier because the heuristics used in the method work for two-class classification, namely: altered and original, but not for the six classes used in our work. Since feature extraction is the main contribution of this method, we replaced heuristics with the above-mentioned classifier for multiclass classification in this work. The classifier consists of two linear layers with "Softmax" [34] as an activation function. During training, it uses "ADAM" [35] as an optimizer and "Categorical Cross Entropy" [36] as the loss function defined in (9).

For validating the enhancement results given by the proposed CM approach, we considered the standard quality measures, namely: naturalness image quality evaluator (NIQE), which measures image quality affected by arbitrary distortion, and BRISQUE, which measures perceptual quality (visual) of images affected by specific distortions. The reason for defining these measures to evaluate the enhancement results is that

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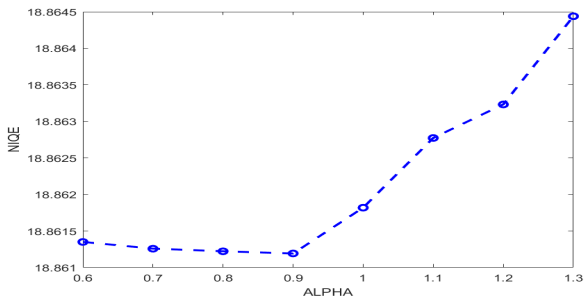


Fig. 8. Illustration to determine the value of alpha experimentally using the NIQE score.

they do not require ground truth, nor the original images for evaluation [44], [45]. In addition, since the handwriting images being considered are affected by both arbitrary distortions caused by forgery operations and specific operations caused by noise and blur, along with high variations in handwriting, NIQE and BRISQUE are suitable measures for judging the quality of enhanced images given by the proposed CM-based approach. It is noted that a low score of NIQE and BRISQUE indicates better quality of the enhanced images. For validating altered handwriting detection, we used well-known measures, namely, the confusion matrix of six classes and two classes, and the average classification rate (ACR), which is the mean of diagonal elements in the confusion matrix.

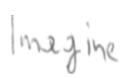
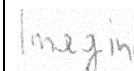
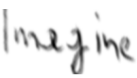
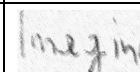
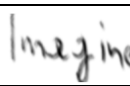
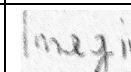
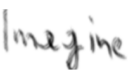
In the proposed CM image enhancement model, the parameter α plays a vital role in achieving better enhancement results. As described in (7), the key parameter α is the power parameter of the proposed CM basis function. The role of the basis function is to represent the image features.

To choose optimal values, which can give better results across datasets, the proposed method chooses 500 random samples from all the datasets, and estimates the NIQE measure for varying alpha values, as shown in Fig. 8. It is observed from Fig. 8 that when the alpha value is 0.9, the proposed method reaches the best NIQE score, which is the lowest value compared to the other NIQE scores of different alpha values. Therefore, the value of alpha is chosen as 0.9 and the same is set for all the experiments in this work. In Fig. 8, the proposed method uses the NIQE measure because it measures image quality affected by arbitrary distortions, unlike the BRISQUE measure, which measures the quality when affected by specific distortions.

To validate the hypothesis and to understand the effectiveness of alpha defined in (7), we present qualitative results by varying the alpha values in Table II for sample images, which are blurred and noisy. It is observed from Table II that as the value of alpha changes, the brightness of the content and the respective quality measure values change. For the value of alpha as 9, the image quality is better in terms of brightness compared to the results of other alpha values. The same is true for NIQE and BRISQUE. This infers that the value of alpha as 9 is feasible for achieving better results.

TABLE II

EFFECT OF DIFFERENT VALUES OF ALPHA FOR BLURRED AND NOISY SAMPLE IMAGES

Parameter	Blurred sample			Noisy sample		
	Image	N	B	Image	N	B
$\alpha = 0$		13.3	45.3		25.1	43.4
$\alpha = 0.6$		13.4	44.5		24.3	43.4
$\alpha = 0.9$		12.5	44.4		24.1	43.2
$\alpha = 1.5$		14.6	44.6		24.9	43.4

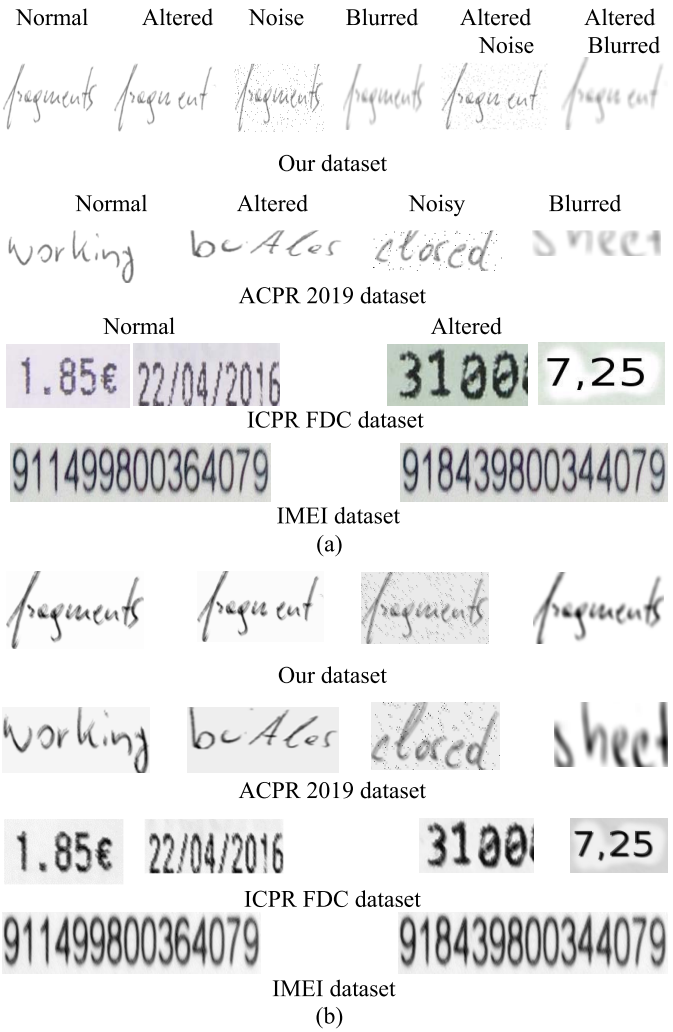


Fig. 9. Examples of the input images and the enhanced images of CMs. (a) Sample input images of different datasets and (b) enhanced images for the corresponding images in (a).

A. Experiments on Enhanced CM Images

Qualitative results of the proposed method for our dataset, the ACPR-2019, ICPR 2018-FDC, and IMEI datasets are shown in Fig. 9. When we look at the input images shown in Fig. 9(a) and the corresponding enhanced images shown in Fig. 9(b), the brightness of finer details, such as edge pixel

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TABLE III

QUALITY MEASURES FOR THE INPUT IMAGES OF OUR DATASET, THE ACPR 2019, ICPR 2018-FDC, AND IMEI DATASETS. (N AND B DENOTE NIQE AND BRISQUE QUALITY MEASURES, RESPECTIVELY, "–"INDICATES THAT THE DATASET DOES NOT INCLUDE A CLASS)

Dataset	Normal		Altered		Noise		Blurred		Altered-Noisy		Altered-Blurred	
	N	B	N	B	N	B	N	B	N	B	N	B
Our	15.4	**45.8**	14.7	46.9	28.2	47.3	14.9	48.0	20.7	46.1	**13.9**	47.6
ACPR 2019	**16.0**	**44.9**	25.0	46.3	25.0	46.3	16.1	47.5	--	-	-	-
Classes	Normal				Altered				--	--	--	--
Measures	N		B		N		B		--	--	--	--
ICPR-FDC	**14.1**		**43.2**		17.7		43.4		--	--	--	-
IMEI	**12.6**		**43.1**		12.8		43.3		--	--	--	-

TABLE IV

QUALITY MEASURES FOR ENHANCED IMAGES OF OUR DATASET

Methods	Normal		Altered		Noise		Blurred		Altered-Noise		Altered-Blurred	
	N	B	N	B	N	B	N	B	N	B	N	B
Proposed	**14.1**	48.3	**13.4**	**46.8**	**24.4**	**43.8**	**13.2**	**46.8**	**21.2**	**44.9**	**12.4**	**47.0**
Chowdhury [43]	14.4	46.5	13.8	47.2	30.3	47.7	13.8	47.2	22.1	45.8	13.6	47.5
Shamasneh [26]	15.3	**45.9**	14.5	47.0	26.6	47.5	14.9	47.9	28.3	50.1	12.7	47.4
Raghunandan [27]	15.5	47.2	14.8	47.8	28.9	46.5	15.3	47.4	21.4	45.6	12.8	48.3

TABLE V

QUALITY MEASURES FOR ENHANCED IMAGES OF THE ACPR 2019 DATASET

Methods	Normal		Altered		Noise		Blurred	
	N	B	N	B	N	B	N	B
Proposed	**15.6**	45.3	17.1	**45.7**	**17.1**	**45.7**	**16.1**	**44.6**
Chowdhury [43]	15.61	**39.05**	**13.97**	47.43	25.12	46.78	16.44	47.50
Shamasneh [26]	15.85	45.13	24.88	46.53	24.88	46.53	16.13	47.45
Raghunandan [27]	16.52	46.22	25.97	45.91	25.97	45.91	16.50	46.96

TABLE VI

QUALITY MEASURES FOR ENHANCED IMAGES OF THE ICPR2018-FDC AND IMEI DATASETS

Methods	ICPR2018-FDC				IMEI			
	Normal		Altered		Normal		Altered	
	N	B	N	B	N	B	N	B
Proposed	**13.75**	**36.60**	15.34	**31.71**	12.61	**41.36**	14.31	42.85
Chowdhury [43]	17.43	42.80	15.48	41.63	13.76	43.18	12.67	42.96
Shamasneh [26]	17.56	43.01	**14.33**	42.20	12.02	43.98	11.66	**12.02**
Raghunandan [27]	14.67	43.02	31.29	43.02	**9.50**	41.93	**9.74**	41.92

increases, are compared to the edge pixels in the input images irrespective of content and application. Therefore, we can conclude that the proposed method is robust, effective, and independent of content and application.

Quantitative results of the proposed and the existing methods for input images of different classes of our dataset, the ACPR-2019, ICPR 2018-FDC, and IMEI datasets, as well as the enhanced images of the respective datasets, are reported in Tables III–VI. Note that ours and the ACPR-2019 datasets provide four classes, namely: normal, altered, noisy, and blurred, while the ICPR 2018-FDC and IMEI datasets provide normal and altered classes. It is observed from Table II that the BRISQUE measure scores low values for normal, and high values for the other classes on all the datasets. This confirms that the images of altered, noisy, blurred, altered-noisy, and altered-blurred classes are suffering from poor quality due to distortion effects.

When we look at the results reported in Tables IV–VI, the proposed method achieves the best NIQE and BRISQUE for almost all the classes across the datasets compared to the existing methods. However, for the IMEI dataset, the performance of the proposed method degrades slightly compared to the existing methods. This is because the images are complex compared to the other datasets, and forgeries are limited to only numerals, unlike many character variations. This shows that the proposed method performs better regardless of dataset, content, and applications. This is because the proposed CMs enhance the finer details in the images, although they are affected by noise and blur. However, since the existing methods focus on specific applications and situations, the methods suffer from inherent limitations for achieving better results on the datasets considered in this work.

TABLE VII

STUDYING EFFECTIVENESS OF THE KEY STEPS OF THE PROPOSED WORK ON OUR DATASET (IN %). (NR: NORMAL, A: ALTERED, N: NOISY, B: BLURRED, AN: ALTERED-NOISY, AB: ALTERED-BLURRED, AND ACR)

Classes	DENN without CM						CM + Stenographic features					
	NR	A	N	B	AN	AB	NR	A	N	B	AN	AB
NR	69.0	30.0	1.0	0.0	0.0	0.0	75.0	25.0	0.0	0.0	0.0	0.0
AL	41.0	58.0	0.0	1.0	0.0	0.0	27.0	73.0	0.0	0.0	0.0	0.0
N	0.0	1.0	70.0	4.0	26.0	0.0	0.0	0.0	80.0	1.0	19.0	0.0
B	0.0	0.0	1.0	66.0	0.0	33.0	0.0	1.0	0.0	77.0	0.0	22.0
AN	0.0	0.0	29.0	0.0	71.0	0.0	0.0	0.0	16.0	0.0	84.0	0.0
AB	0.0	0.0	0.0	32.0	0.0	68.0	0.0	0.0	0.0	30.0	3.0	67.0
ACR	67.00						76.00					

TABLE VIII

COMPARING THE KEY STEPS WITH THE PROPOSED METHOD ON OUR DATASET (IN %)

Classes	CM + Photo Montage						Proposed (CM+Stenographic + Photo Montage)					
	NR	A	N	B	AN	AB	NR	A	N	B	AN	AB
NR	62.0	21.0	0.0	9.0	0.0	8.0	87	13	0.0	0.0	0.0	0.0
A	7.5	60.0	0.0	15.0	4.0	3.5	21.0	79.0	0.0	0.0	0.0	0.0
N	0.0	0.0	78.0	1.0	21.0	0.0	0.0	0.0	84.0	0.0	16.0	0.0
B	0	0.0	2.0	76.0	2.0	20.0	0.0	0.0	0.0	86	0.0	14
AN	0.0	0.0	25.0	0.0	75.0	0.0	0.0	0.0	12.0	0.0	88.0	0.0
AB	0.0	0.0	0.0	20.0	0.0	80.0	0.0	0.0	0.0	10	0.0	90
ACR	71.84						**85.67**					

B. Ablation Study

The key steps of the proposed method are CMs for enhancing the finer details in the input images, a DENN which involves two sub-steps, namely: stenographic residual and photograph montage for feature extraction. To analyze the contribution of the key steps, we estimate the confusion matrix of classification, and we calculate the ACR for our dataset with different combinations; the results are reported in Tables VII and VIII. The results using only DENN (which

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TABLE IX

CONFUSION MATRIX AND ACR OF THE EXISTING METHODS [21], [23] ON OUR DATASET (IN %)

Classes	Shivakumara [19]						Nandanwar et al [21]					
	NR	A	N	B	AN	AB	NR	A	N	B	AN	AB
NR	61.0	20.0	5.0	10.0	0.0	4.0	60.0	0.0	16.0	5.0	12.0	7.0
A	14.0	44.0	15.0	1.0	16.0	10.0	11.0	62.0	5.0	7.0	12.0	3.0
N	0.0	0.0	74.0	3.0	12.0	1.0	0.0	0.0	64.0	06.0	29.0	1.0
B	0.0	1.0	4.0	69.0	2.0	24.0	0.0	5.0	1.0	51.0	4.0	39.0
AN	0.0	0.0	25.0	20.0	50.0	15.0	0.0	0.0	20.0	3.0	71.0	6.0
AB	0.0	0.0	0.0	20.0	7.0	73.0	0.0	0.0	0.0	33.0	2.0	65.0
ACR	61.83						62.17					

TABLE X

CONFUSION MATRIX AND ACR OF THE EXISTING METHODS [12], [15], [22] ON OUR DATASET (IN %)

Classes	Megahed et.al [22]						Kundu et al. [15]						Wang et.al. [12]					
	NR	A	N	B	AN	AB	NR	A	N	B	AN	AB	NR	A	N	B	AN	AB
NR	80.0	11.0	0.0	5	0.0	4.0	52.0	45.0	0.0	2.0	0.0	1.0	52.0	20.0	0.0	11.0	0.0	17.0
A	40.0	39.0	0.0	8.0	0.0	13.0	48.0	48.0	0.0	2.0	0.0	2.0	21.0	46.0	0.0	14.0	0.0	19.0
N	9.0	0.0	68.0	0.0	23.0	0.0	0.0	0/0	85.0	5.0	15.0	5.0	5.0	4.0	66.0	0.0	25.0	0.0
B	13.0	0.0	0.0	71.0	0.0	16.0	2.0	0.0	2.0	57.0	16.0	23.0	0.0	10.0	0.0	64.0	0.0	26.0
AN	15.0	0.0	18.0	0.0	67.0	0.0	0.0	0.0	15.0	5.0	75.0	5.0	0.0	0.0	34.0	7.0	56.0	3.0
AB	12.0	0.0	0.0	13.0	0.0	75.0	0.0	0.0	5.0	22.0	5.0	68.0	0.0	0.0	0.0	37.0	0.0	63.0
ACR	66.67						64.17						57.83					

considers input images directly for classification without enhancement) shows that the ACR is lower than the proposed method. Therefore, we can confirm that CMs are effective and contribute to achieving better results for classification. In the same way, we compute the confusion matrix and the ACR for the features extracted using stenographic residuals and a Photograph Montage from both input and enhanced images, as reported in Table VII. It is noted from Tables VII and VIII that both the features contribute almost equally, as ACR is higher than DENN and lower than the proposed method to achieve better classification rates. However, when we compare the contributions of stenographic and photograph montage features, the Stenographic features contribute in a greater way than the photograph montage. This makes sense because Stenographic features are more powerful than the conventional features as they extract high-level context features, while photograph montage extracts low-level features. In summary, we can conclude that the key steps are essential in achieving the better results for the proposed method, as it scores the overall highest ACR compared to each of the individual steps.

C. Evaluating the Proposed Model for Altered Handwriting Detection

Tables IX and X report the results of the existing methods on our dataset. In comparison, our proposed method is better in terms of the ACR. The method [22] is better than all other existing methods listed in Tables IX and X. This is because the method [22] extracts feature at the character level rather than at the word level, in contrast to the other methods, providing more local information for detecting character-level forgeries. However, the performance of the method in [22] is lower than the proposed method. The reason is that the features of the method in [22] are not robust to noise and blur. It can be observed from the results on the ACPR

TABLE XI

CONFUSION MATRIX AND ACR OF THE PROPOSED AND THE EXISTING METHODS ON THE ACPR 2019 DATASET (IN %). (NR: NORMAL, A: ALTERED, N: NOISY, B: BLURRED, AND ACR)

Classes	Proposed Method				Kundu et al. [15]				Wang et.al. [12]			
	NR	A	N	B	NR	A	N	B	NR	A	N	B
NR	97.5	2.5	0.0	0.0	70.0	20.0	10.0	0.0	57.8	25.0	9.4	7.8
A	5.0	95.0	0.0	0.0	4.8	85.7	0.0	9.5	25.0	71.4	1.8	1.8
N	0.0	0.0	100.0	0.0	0	0.0	90.0	10.0	14.3	8.0	76.2	1.5
B	0.0	0.0	0.0	100.0	15.8	15.8	5.2	63.2	9.0	1.8	11.0	78.2
ACR	**98.1**				77.5				70.1			

TABLE XII

CONFUSION MATRIX AND ACR OF THE EXISTING METHODS ON THE ACPR 2019 DATASET (IN %)

Classes	Megahed et.al [22]				Nandanwar et al [23]				Shivakumara [21]			
	NR	A	N	B	NR	A	N	B	NR	A	N	B
NR	77.5	20.5	0.0	2.5	65.0	22.5	5.0	7.5	77.0	23.0	0.0	0.0
A	57.5	32.5	0.0	10.0	27.5	67.5	0.0	5.0	40.0	50.0	0.0	0.0
N	22.5	0.0	75.0	2.5	0	0.0	95.0	5.0	0.0	0.0	90.0	10.0
B	22.5	0.0	0.0	77.5	2.5	2.5	5.0	90.0	22.0	0.0	0.0	78.0
ACR	65.6				79.37				73.75			

TABLE XIII

CONFUSION MATRIX AND ACR OF THE PROPOSED AND THE EXISTING METHODS ON THE ICPR 2018-FDC DATASET (IN %)

Methods	Proposed method		Kundu et. al. [15]		Wang et al. [12]		Shivakumara et al. [21]		Megahed et al. [22]		Nandanwar et al.[23]	
Classes	NR	A	NR	A	NR	A	NR	A	NR	A	NR	A
NR	98.3	1.7	90.0	10.0	84.6	15.4	92.0	8.0	84.3	15.6	92.0	8.0
A	3.2	96.8	72.5	27.5	10.7	89.3	49.4	50.6	28.2	71.7	25.0	75.0
ACR	**97.55**		78.3		86.9		71.3		78.07		83.5	

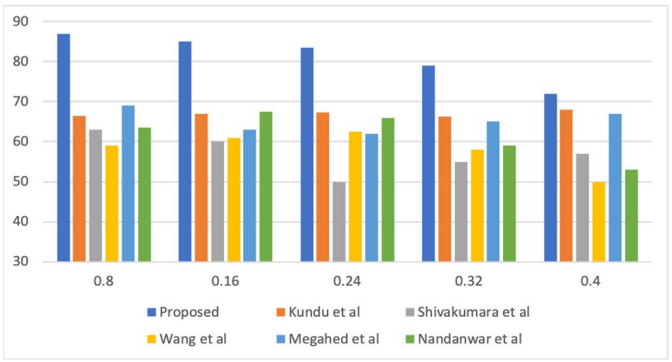
2019 dataset reported in Tables XI and XII that the method in [23] achieves the best performance. This is because this method is robust to forgeries at the word and character levels, like our proposed method. But the method in [23] is not robust to degradations and hence it reports poor results compared to our proposed method. Note also that our method performs better on the ACPR 2019 dataset than on our own dataset. This is because our dataset contains character-level forgeries which are harder to detect, while the ACPR 2019 dataset contains word-level forgeries. The reason for the poor results of the existing methods compared to the proposed method is that the objective of the existing methods was to consider a two-class classification problem but not for performing six-class classification.

For the ICPR 2018-FDC and IMEI datasets, the proposed method is better than the existing methods in terms of ACR as reported in Tables XIII and XIV, respectively. However, it is noted that although IMEI is a two-class classification problem, the proposed method scores slightly lower results because this dataset contains more complex images, and the forgeries are limited to numerals. However, for all of the other datasets, including our own dataset, the proposed method scores consistent results. Interestingly, the method in [22] is better than the other existing methods on our dataset while the method in [23] is better than all the other existing methods

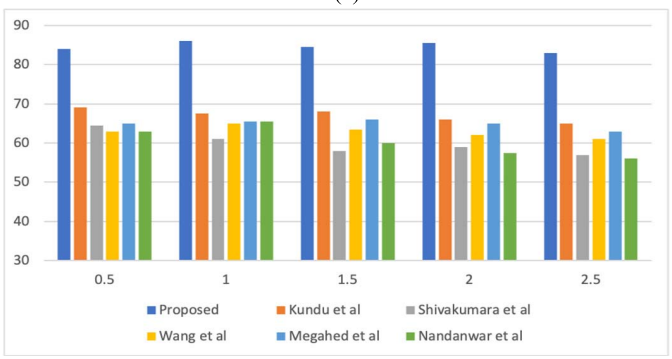
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TABLE XIV
CONFUSION MATRIX AND ACR OF THE PROPOSED AND THE EXISTING METHODS ON THE IMEI DATASET (IN %)

Methods	Proposed method		Kundu et. al. [15]		Wang et al. [12]		Shivakumara et. al. [21]		Megahed et al. [22]		Nandanwar et al. [23]	
Classes	NR	A	NR	A	NR	A	NR	A	NR	A	NR	A
NRl	95.2	4.8	57.8	42.2	83.2	16.8	82.2	17.8	67.6	32.4	78.0	22.0
A	18.2	81.8	41.8	58.2	25.6	74.4	18	82	38.2	61.8	18.0	82.0
ACR	**88.5**		58.0		78.8		82.1		64.7		80.0	



(a)



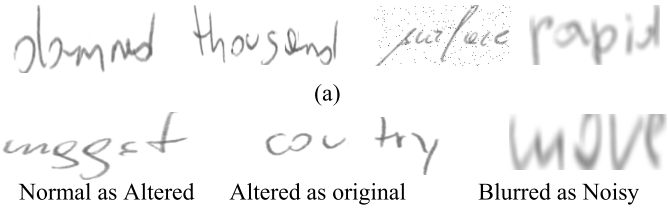
(b)

Fig. 10. Performance of the proposed and the existing methods on images affected by different levels of noise and blur on our dataset. Increasing (a) noise density and (b) blur level.

for ACPR 2019. However, both the methods deliver lower results than Wang *et al.*'s [12] method for the ICPR 2018-FDC and IMEI datasets. The reason is that the method in [22] is developed for detecting forged handwriting and the method in [23] is sensitive to degradations, while Wang *et al.*'s [12] method is developed for forgery detection in printed document images such as the images in ICPR 2018-FDC and IMEI. For the IMEI dataset, Shivakumara *et al.*'s [21] method gives better results than the other two methods because it was developed specifically to detect forged IMEI numbers.

In summary, for a range of different datasets, our proposed method achieves the best overall results in terms of ACR. This shows that the method works well for detecting forgeries in handwriting images, including in situations where there are noise and blur effects.

To test the ability of the proposed and existing methods on whether they can withstand the effects of different noise and blur levels, we conducted experiments by increasing Gaussian noise and blur levels as shown in Fig. 10(a) and Fig. 10(b),



(a)

Normal as Altered Altered as original Blurred as Noisy

(b)

Fig. 11. Examples of successful and unsuccessful classification by the proposed method. (a) Successfully classified samples and (b) unsuccessfully classified samples.

respectively. In this experiment, noise and blur are added to the images of noisy and blurred classes. The ACR of all the classes is computed for measuring the performance of the methods. It is observed from Fig. 10(a) and (b) that our method and Kundu *et al.*'s [15] methods can sustain the same performance up to certain levels, while other existing methods do not. This shows that our approach has the ability to cope with the effect of dense noise and blur. Since the existing methods are not robust to noise, they fail to obtain consistent results when the noise and blur levels increase. However, the performance of the methods decreases after certain levels of noise and blur. This is because images lose vital information and hence the extracted features overlap with the features of other classes.

When abrupt changes caused by forgery operations are small, the method misclassifies images, such as the examples shown in Fig. 11, where we can see both successful and unsuccessful images. Since the dataset being considered is comprised of handwriting, speed and writing force may introduce distortions and create problems for correct classification. Therefore, there is scope for studying the effects of speed, writing force, and age of people for altered text detection in handwriting document images in noisy and blurred environments.

D. Discussion

In order to show the proposed method can be used for many applications, we have used a handwriting dataset containing six classes (our dataset), a dataset of four classes (ACPR 2019 dataset), and a benchmark dataset (ICPR 2018 FDC), which is comprised of tampered receipts, which is in contrast to ours and the ACPR 2019 dataset. In addition, the IMEI dataset is considered, which is comprised of IMEI number images but not plain documents. This shows that the proposed method is independent of applications, and it works well irrespective of content such as handwriting images, printed text images, and text images with complex backgrounds. It is evident from the results on all the datasets that the proposed method outperforms the existing methods for different datasets in terms of ACR. Similarly, the experiments on six, four, and two classes on different datasets, with the results of existing methods, demonstrate that the proposed method does not depend on the number of classes. Furthermore, the results of the proposed and existing methods on images affected by noise and blur illustrate that the proposed method is invariant to noise and blur for altered text detection in the images. Overall,

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the proposed method is robust across different applications, datasets, the number of classes, noise, and blur.

However, when the proposed method does not find inconsistent patterns created by forgery operations, it misclassifies forged images into other classes. This is possible when forgery operations create only a little distortion, at the same time, noise and blur levels increase. It is true that due to free-style writing, there is a chance of generating poor quality because of variations in the speed of writing, writing force, paper, ink, age, gender, etc. This leads to confusion for the classification of forged images. Sometimes, there is a chance of introducing both noise and blur in the original or forged images. In this case, the number of classes and complexities increase. This challenge is beyond the scope of the proposed work, and thus the scope of the proposed work does not include the class of altered-noisy-blurred. These are the limitations of the proposed work, but this will be the target of our future work. In addition, future research will include a study on the effect of age, gender, and online handwriting for forgery detection in document images.

V. Conclusion and Future Work

In this work, we have proposed a novel method for altered handwriting detection in noisy and blurry environments based on CMs, and a DENN. In order to preserve structures of characters in handwriting images, irrespective of noise, blur, and large handwriting variations, we have introduced a novel CM model involving Fractional calculus and geometric moments. The proposed CMs enhance the finer details such as edges in images of six classes, namely normal, altered, noisy, blurred, altered-noisy, and altered-blurred. To extract edge patterns in all the images of all the classes, we have proposed a new deep architecture that combines features extracted from stenographic residuals of the input and the CM images, and a photograph montage. Experimental results on different datasets, such as our dataset, the standard ACPR-2019 dataset, the ICPR 2018-FDC, and the IMEI datasets demonstrate that the proposed CMs and the altered detection approach outperform the existing methods. When the single image affected by multiple forgeries (copy-paste and insertion), the proposed method may not be effective. This will be considered as part of our future work.

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